

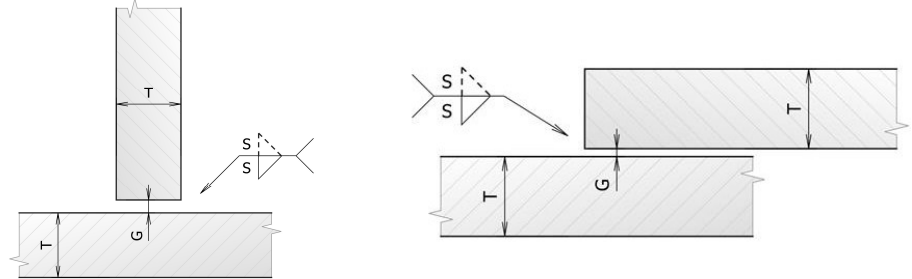
| JSS/POLAR POCKET WPS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                            |                                          |                                                                             |                   |                                | Pocket-POLAR-01-M                                                                                                               |                                            | 1                                                                                        | April 27, 2026        |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------|-------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------|-----------------------|-------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                            |                                          |                                                                             |                   |                                | WPS No.                                                                                                                         |                                            | Rev.                                                                                     | Date                  |                               |
| <div>Seaspan Vancouver Shipyards Co. Ltd.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                            |                                          |                                                                             |                   |                                | Applicable Standard(s)                                                                                                          |                                            | Lloyds Register - Rules for the Manufacture, Testing and Certification of Materials 2023 |                       |                               |
| Process/Mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                            | Electrode (Wire) Classification          |                                                                             |                   |                                | Brand Name(s)                                                                                                                   |                                            | Manufacturer(s)                                                                          |                       |                               |
| FCAW/Semi-Auto(Hand)<br>FCAW/Auto(Moggy)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                            | AWS A5.20 E71T-1M/9M-J   LR Grade: 4Y40S |                                                                             |                   |                                | Dual Shield Prime 71 LT H4                                                                                                      |                                            | ESAB                                                                                     |                       |                               |
| Material Designation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Base material 1                                                                            |                                          | Base material 2                                                             |                   | Min. Preheat / Interpass Temp. | 5°C for 3 ≤ T ≤ 20mm<br>10°C for 20 < T ≤ 40mm<br>65°C for 40mm < T ≤ 60mm<br>110°C for T > 60<br>(See Note 7, Note 8 & Note 9) |                                            |                                                                                          |                       |                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | EH36, DH36 and all lower grades excluding A,B, D and E<br>(Note 4 & Note 5)                |                                          | EH36, DH36 and all lower grades excluding A,B, D and E<br>(Note 4 & Note 5) |                   |                                |                                                                                                                                 |                                            |                                                                                          |                       |                               |
| Thickness or Dia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3 to 100 mm                                                                                |                                          | 3 to 100 mm                                                                 |                   | Max. Interpass Temp.           | 180°C                                                                                                                           |                                            |                                                                                          |                       |                               |
| Nominal Pipe Size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 500mm and above                                                                            |                                          | 500mm and above                                                             |                   | PWHT                           | N/A                                                                                                                             |                                            |                                                                                          |                       |                               |
| Welding Position                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | * All positions excluding Vertical down<br>* Ceramic/Steel Backing: 1G, 2G and 3G-up ONLY. |                                          |                                                                             |                   | Joint Design                   | CJP & PJP Groove Single or Double Bevel / Vee, T- K- Y Joint, Fillet                                                            |                                            |                                                                                          |                       |                               |
| TYPICAL JOINT CONFIGURATIONS ON NEXT PAGE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                            |                                          |                                                                             |                   |                                |                                                                                                                                 |                                            |                                                                                          |                       |                               |
| COMPLETE JOINT PENETRATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                            | Welding Layer                            |                                                                             | JOINT TYPE        | Back Purge                     | N/A                                                                                                                             |                                            | Contact Tip to Work Distance                                                             |                       | 9.5 - 20 mm (Target 15 mm)    |
| <div><div>Back-gouged to sound metal</div><div>Welded onto backing</div><div>Welded from one side without backing</div><div>Welded both sides w/o back-gouging</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                            | Single/Multi Layer                       |                                                                             | BUTT              | Backing type                   | If used, Ceramic/Steel                                                                                                          |                                            | Interpass Cleaning                                                                       |                       | Grinding and Wire Wheel       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                            | One/Two side                             |                                                                             | CORNER            | Welding Technique              | Stringer/Slight Weave                                                                                                           |                                            | Shielding Gas                                                                            |                       | 75% Argon / 25% CO2           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                            | Gun travel angle <sup>3</sup>            |                                                                             | LAP               | Max. Bead Width                | Multi-Run: 18 mm<br>Single-Run: 15 mm                                                                                           |                                            | Gas Flow                                                                                 |                       | 16-25 LPM                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                            | Pull w/ slight push on Vu                |                                                                             | TEE               | No. of electrodes              |                                                                                                                                 | 1                                          |                                                                                          |                       | 34-53 CFH                     |
| Method of steel preparation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                            | Oxy fuel/Plasma cut   Grinding   Milling |                                                                             | EDGE              |                                |                                                                                                                                 |                                            |                                                                                          |                       |                               |
| BM Thickness, T(mm)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Layers / Passes                                                                            | Electrode Size (mm)                      | Welding Process                                                             | Power Mode        | Consumable                     | Amperage (A)                                                                                                                    | Voltage (V)                                | WFS (IPM)                                                                                | Travel Speed (mm/min) | Heat Input <sup>1</sup> kJ/mm |
| 3 ≤ T ≤ 24                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Single-Pass Fillet                                                                         | 1.2 - 1.6                                | FCAW (Semi-Auto)                                                            | DCEP              | E71T-1M/9M                     | 180 - 300                                                                                                                       | 22 - 30                                    | 170 - 500                                                                                | 147 - 196             | 1.1 - 2.2                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                            |                                          | FCAW (Auto-Moggy)                                                           | DCEP              | E71T-1M/9M                     | 200 - 350                                                                                                                       | 24 - 32                                    | 170 - 500                                                                                | 210 - 288             | 1.0 - 2.4                     |
| 3 ≤ T ≤ 24                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Root                                                                                       | 1.2 - 1.4                                | FCAW (Semi-Auto)                                                            | DCEP              | E71T-1M/9M                     | 140 - 250                                                                                                                       | 22 – 28                                    | 170 - 300                                                                                | 55 - 130              | 1.5 - 2.9                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Hot/Fill/Cap Multi-Run Fillet                                                              | 1.2 - 1.6                                | FCAW (Semi-Auto)                                                            | DCEP              | E71T-1M/9M                     | 180 - 485                                                                                                                       | 22 - 29                                    | 170 - 500                                                                                | 108 - 625             | 0.5 - 1.6                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                            |                                          | FCAW (Auto-Moggy)                                                           | DCEP              | E71T-1M/9M                     | 180 - 420                                                                                                                       | 22 - 29                                    | 170 - 500                                                                                | 75 - 660              | 0.5 - 1.6                     |
| 24 < T ≤ 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Root                                                                                       | 1.2 - 1.4                                | FCAW (Semi-Auto)                                                            | DCEP              | E71T-1M/9M                     | 140 - 250                                                                                                                       | 22 - 30                                    | 170 - 300                                                                                | 55 - 130              | 1.7 - 3.9                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Hot/Fill/Cap Multi-Run Fillet                                                              | 1.2 - 1.6                                | FCAW (Semi-Auto)                                                            | DCEP              | E71T-1M/9M                     | 180 - 485                                                                                                                       | 20 - 30                                    | 170 - 500                                                                                | 108 - 625             | 0.5 - 2.6                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                            |                                          | FCAW (Auto-Moggy)                                                           | DCEP              | E71T-1M/9M                     | 180 - 420                                                                                                                       | 22 - 30                                    | 170 - 500                                                                                | 75 - 660              | 0.7 - 3.9                     |
| <div><div>Note 1:</div><div>Heat Input (kJ/mm) = [V x A x 60] / [Travel Speed (mm/min) x 1000]</div><div>Note 2:</div><div>Clean joint and adjacent areas to bright metal. Remove paint, primer, rust, scale, moisture, and contaminants. Wire brush between passes.</div><div>Note 3:</div><div>Travel angle 5–10° pull. For vertical up, use slight push.</div><div>Note 4:</div><div>Qualified for all delivery conditions except quenched and tempered steel.</div><div>Note 5:</div><div>Welding Grade A, B, D, E hull steel with Dual Shield Prime 71 LT requires Lloyd's special agreement for LR Class vessels.</div><div>Note 6:</div><div>For fillet welds, if root gap G &gt; 3 mm, increase fillet leg size by (G – 2 mm). If G &gt; 5 mm, consult Welding Engineering.</div><div>Note 7:</div><div>See Preheat &amp; Interpass WI (SQMS-VSY-WEQ-P002) for other base materials.</div><div>Note 8:</div><div>If visible moisture is present on the surface, preheat the joint to a minimum of 20°C prior to welding.</div><div>Note 9:</div><div>For ceramic backing up to 40 mm thickness, the recommended minimum preheat temperature for root and hot passes is 50°C.</div><div>Note 10:</div><div>For the root pass on the ceramic backing, it is recommended to use 1.2mm (0.045") wire size with max. 200 amps.</div><div>Note 11:</div><div>For ceramic backing, rectangular groove tiles recommended. If round ceramic is used, follow with GTSM.</div><div>Note 12:</div><div>If more information from the welding procedure is required, please refer to Weldeye or consult Welding Engineering.</div></div> |                                                                                            |                                          |                                                                             |                   |                                |                                                                                                                                 |                                            |                                                                                          |                       |                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                            |                                          |                                                                             |                   |                                |                                                                                                                                 |                                            | Reference WPS No.                                                                        |                       |                               |
| Bug-O Weave Settings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Dwell (L&R): 0 - 1.0                                                                       |                                          |                                                                             | Weave Width: 0-18 |                                |                                                                                                                                 | FC-CS-G-03 (Rev. 4)<br>FC-CS-F-03 (Rev. 0) |                                                                                          |                       |                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Weave Pattern No.:<br>#1 for 2G / #1 & #3 for other                                        |                                          |                                                                             | Weave Speed: 0-50 |                                |                                                                                                                                 |                                            |                                                                                          |                       |                               |

# TYPICAL JOINT CONFIGURATIONS

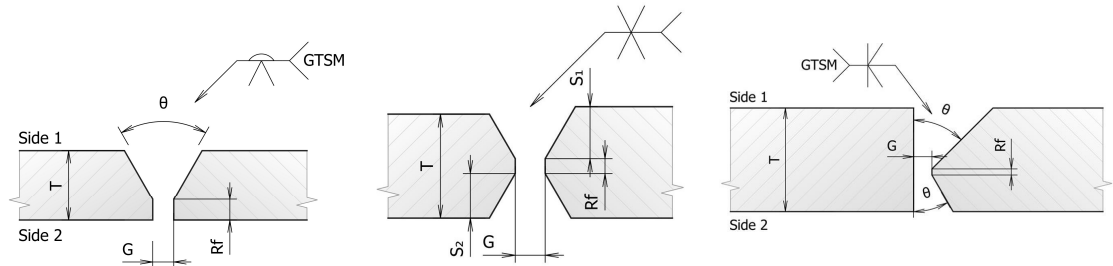
## Fillet Weld

Fillet Leg Size (Multi-Run) = 4.2 to 140 mm

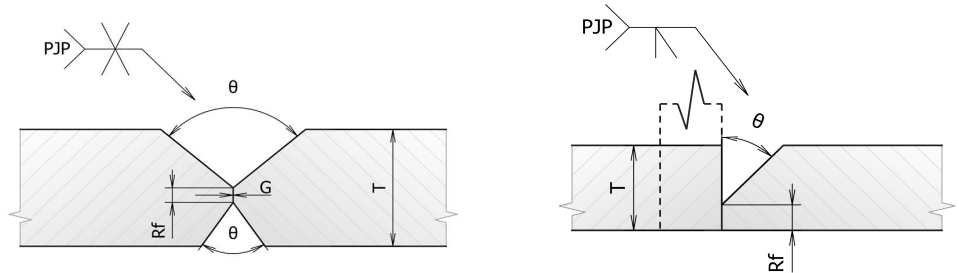
Fillet Leg Size (Single-Run) = 5 to 12 mm

Fillet Root Gap = 0 to 2 mm  
(See Note 6)

## Back-gouged to Sound Metal (GTSM)

Root Gap (**G**) = 0mm to 3 mm for T up to 8 mmRoot Gap (**G**) = 0mm to 4.5 mm for T over 8 mmRoot Face (**Rf**) = 3mm minimumGroove angle ( $\theta$ ) = 40-70°

## Partial Joint Penetration (PJP)

Root Gap (**G**) = 0 - 1mmRoot Face (**Rf**) = 2mm minimumGroove angle ( $\theta$ ) = 40-70°

## Welded Onto Backing

Root Gap for Ceramic (**G**) = Target 4 to 7 mm  
(max 1.5xT or 11mm, whichever smaller, is acceptable)Root Gap for Steel (**G**) = 5 mm minimumRoot Face for Ceramic/Steel (**Rf**) = 0 to 1 mmGroove angle ( $\theta$ ) = 40 - 70°

See Note 10, Note 11

